

Md Arifuzzaman Swapnil

Backend AI Engineer

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SUMMARY

Backend AI Engineer specialising in production RAG pipelines and LLM-powered applications. I design and ship scalable Python backends — REST APIs, AI agents, and intelligent chatbots — for live SaaS products across proptech and construction tech. Recent work includes RAG systems that answer questions over contracts and property documents with clause-level cited references, GPT-4o Vision document analysis, and end-to-end AI automation workflows. Comfortable across the full stack, with a focus on turning LLMs into reliable, production-ready features.

EDUCATION

B.Sc. in Computer Science & Engineering

2021 – 2025

Bangladesh University of Business and Technology (BUBT) CGPA: 3.92 / 4.00

Higher Secondary Certificate (HSC)

2018 – 2020

Government Science College GPA: 5.00 / 5.00

Secondary School Certificate (SSC)

2017 – 2018

Shaheed Police Smriti College GPA: 5.00 / 5.00

WORK EXPERIENCE

Backend AI Engineer

Feb 2026 – Present

[Intellect AI](#)

- Build and scale Python backend services and REST APIs powering live SaaS products (HomePlus, TradePilot, BaseLinq).
- Design and implement Retrieval-Augmented Generation (RAG) pipelines that answer questions over construction contracts and property documents with clause-level cited references.
- Develop custom chat models and LLM-powered AI agents, and engineer backend caching and scalability strategies (e.g. Redis) for high-traffic endpoints.
- Build intelligent chatbots and AI automation workflows integrated directly into customer-facing products.

Full-Stack Python Web Developer | AI Engineer

Jan 2025 – Feb 2026

[Abedin Tech](#)

- Built and maintained full-stack Python web applications and scalable backend services end to end.
- Developed Python REST APIs, automation scripts, and background job pipelines (Celery).
- Created AI automations, AI agents, chatbots, and AI calling systems integrating GPT-4o, Claude, and Gemini.
- Integrated Stripe payments with a token-based billing system and built responsive React + TypeScript frontends for customer-facing SaaS products.

Teaching Assistant

Oct 2024 – Dec 2025

[Bangladesh University of Business and Technology \(BUBT\)](#)

- Delivered lectures and hands-on lab sessions in Object-Oriented Programming (C++), Artificial Intelligence, Machine Learning, and Neural Networks.
- Mentored undergraduate students through assignments, projects, and exam preparation.

TECHNICAL SKILLS

AI & LLM Engineering: RAG pipelines, LLM application development (OpenAI GPT-4o, Anthropic Claude, Google Gemini), AI agents, chatbots, AI calling systems, prompt engineering, embeddings & vector search.

Backend & APIs: Python (Advanced), Django, Django REST Framework, FastAPI, Flask, Celery, JWT auth, REST API design, scalable backend architecture, caching (Redis).

Frontend: React, Next.js, TypeScript, HTML5, CSS3, Tailwind CSS.

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Firebase.

ML & Data: Dataset preparation, EDA, data preprocessing, model training.

Automation & Tools: n8n, Zapier, Make, API automation, Git, VS Code, Jupyter Notebook, Google Colab, Kaggle.

Languages: English (professional), Bengali (native).

PROFESSIONAL PROJECTS

BaseLinq AI — Construction Project Management Platform *(Intellect AI)*

[Project Link](#)

Full-stack construction project management platform (React + TypeScript frontend, Django REST + JWT backend) for managing projects, tasks, and contractual workflows such as Variation Orders, RFIs, Site Instructions, and Delay Claims — with automated numbering, financial calculations, and approval flows. Added role-based access control, cost tracking, web-push notifications, and AI-powered meeting transcription/summarisation (Recall.ai + Claude). Built Linq AI, a RAG-based chatbot that answers questions on project documents and contracts with cited, clause-level references.

Sellanto — AI-Powered Social Media Automation Platform *(Abedin Tech)*

[Project Link](#)

Full-stack SaaS platform that automates AI-driven content creation, scheduling, and publishing across 9 social networks, integrating OpenAI GPT-4o, Claude, and Gemini for caption, image, and video generation plus a RAG-based Messenger chatbot. Built a Django REST API (JWT, Celery scheduling) with a React 19 + TypeScript frontend, a “Brand DNA” engine, engagement analytics, and dual billing (Stripe + token economy) with a full admin panel.

HomePlus & TradePilot — Connecting Homeowners with Trusted Tradespeople *(Intellect AI)*

[Project Link](#)

HomePlus:

A React + Django/DRF web application enabling homeowners to request, manage, and track home improvement jobs, with quote generation, job categorisation, and document uploads. Integrated an OpenAI GPT-4o Vision pipeline that reads property documents and produces AI-driven EPC energy ratings, including estimates for non-EPC files.

TradePilot:

[Project Link](#)

A companion platform for tradespeople to receive, quote, and manage incoming jobs in real time, sharing a unified backend and reusable component library with HomePlus over Django REST APIs — streamlining the end-to-end flow from customer request to tradesperson quote.

TechStyles — AI-Powered Design Studio & Procurement Platform *(Intellect AI)*

[Project Link](#)

Full-stack platform managing the complete interior-design lifecycle — from project phases and room-level planning to product procurement, contractor coordination, and client approvals. Built dedicated client and contractor portals, a built-in CRM, and finance modules integrated with Xero for accounting and AWS S3 for document storage. Powered by a Django REST Framework backend and a responsive Next.js (TypeScript, Tailwind CSS) frontend, with OpenAI-driven AI features, time tracking, meetings, and reporting.

AWARDS & ACHIEVEMENTS

ITEE (Oct 2024): Ranked 30th in Bangladesh in the Information Technology Engineers Examination (ITEE), administered by IPA, Japan — one of the most competitive IT certification exams in the country.

Competitive Programming: 5th Place, Senior Division, Intra-University Competitive Programming Contest, BUBT 2024.

CERTIFICATIONS

Fullstack Laravel Developer – InteractiveCares

Software Quality Controller – FastFlowUp

Natural Language Processing – Innovative Skills BD Ltd.